

Probability Models for Tennis Scoring Systems

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SUMMARY

Models are developed for each of the common tennis tiebreaker scoring systems based on a fixed probability for each player to win a point on his serve. The models are used to compare each scoring system. Assumptions about and predictions from the models are tested using data from actual tennis matches.

Keywords: Tennis scoring systems; Markov chains; Goodness of fit

1. Introduction

Although the game of tennis dates as far back as the 12th century, the code of scoring as we essentially know it today was not formulated until 1876. The method of scoring by 15s is certainly medieval in origin. The number 60 had a special significance in the early centuries and thus the entire game was awarded a value of 60 points and divided into four equal parts each worth 15 points. Through the years the score of 45 became abbreviated to just 40. It was not until the early 1870s that the modern game of lawn tennis came into existence, primarily because not until then was anyone able to make a rubber ball that would bounce on grass. Soon thereafter, the All England Croquet Club, faced with deteriorating finances, proposed that one of its lawn courts at Wimbledon be converted to tennis and that the club sponsor a tennis tournament complete with a silver challenge cup worth 25 guineas. Since a competition with a monetary prize required a firm code of rules, a committee was appointed to formulate such a code, which has essentially remained unchanged since that time.

In the inaugural Wimbledon Championship in 1877, the first player to win 6 games won the set. If the score reached 5 games all, then the winner of the next game won the set. The first player to win 3 sets won the match. It quickly became apparent, however, that the server of the 11th game of any set had an unfair advantage over his opponent and thus the idea of the advantage set, in which a player must win by at least a two-game margin, was introduced in the 1878 Championships.

The next change in scoring did not occur until 1970 with the unofficial introduction of the tiebreaker at the Philadelphia Open Championships. That same year the United States Lawn Tennis Association gained official approval for use of a 9 point tiebreaker at the United States Open Championships at Forest Hills. The following

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year Wimbledon introduced the 12 point tiebreaker when a set reached a score of 8 games all, except in the fifth set for men and the third set for women. Later they began using the tiebreaker at a score of 6 games all. Many professional tournaments now use the 12 point tiebreaker scoring system.

James Van Alen, one of the original proponents for the use of tiebreakers, has devised several other schemes for scoring in tennis, commonly known as Van Alen Streamlined Scoring Systems (VASSS). In 1958 he introduced the VASSS Single Point in which the first player to win 31 points wins the set, with a 9 point tiebreaker played if the score reaches 30 points all. The serving sequence is similar to ping-pong in that each player serves for 5 points with service then rotating to the other player. In the VASSS No-Ad system, the first player to win 4 points wins the game, the 7th point of a game becoming a game point for each player. If a No-Ad set reaches 6 games all, a 9 point tiebreaker is played. While Single Point scoring has never caught on, the No-Ad system is authorised by the United States Tennis Association for tournaments held at the Sectional Championship level and below, and for consolation matches in any tournament. Several college conferences also use No-Ad scoring during competitive matches.

In this paper we will discuss simple probability models for each of the scoring systems in current use. In their classic text *Finite Markov Chains*, Kemeny and Snell modelled a game in tennis as a preliminary process followed by a random walk of 5 states with two end absorbing states, with the preliminary process providing the initial probabilities for each of the 5 states of the random walk. Their model assumed a fixed probability p for player A to win any given point, regardless of which player held the serve in the game. For our models we will assume that player A has a probability p_A of winning a point *when player A is serving* and that player B has a probability p_B of winning a point *when player B is serving*. More generally, consider a competition consisting of a sequence of contests played between two players (or teams) A and B where the winner of each contest receives one point. Victory is determined by one of the following methods:

- (1) the first player to obtain at least n points with an advantage of at least 2 points or
- (2) the first player to obtain $n + 1$ points out of $2n + 1$ contests played.

We assume that there are two types of contests: A -contests, in which A has some particular 'advantage' (for example, the serve in tennis or the home team in baseball), and B -contests, in which B has the 'advantage'. (Miles (1984) calls these *bipoints*.) We further assume that all contests are mutually independent and that A ($[B]$) wins each A -contest ($[B]$ -contest) with fixed probability p_A ($[p_B]$).

Theorem. (a) For competitions using scoring system (1) in which A -contests and B -contests alternate, the probability distribution of scores is independent of the type of initial contest.

(b) For competitions using scoring system (2) in which there are played at most n of one type of contest and $n + 1$ of the other, the probability distribution of scores is independent of the *order* in which the contests are played.

Proof. (a) First observe that the distribution of the score after an even number of contests does not depend on which type of contest was played first. Player A wins

the competition with a score of n to k for $k \leq n - 2$ if and only if in a sequence of the first $2n - 2$ contests, the score stands in A 's favour (whether or not all such contests actually needed to be played). By the first remark, therefore, the probability distribution of such a result is independent of the type of the initial contest. If the competition is not decided by such a score, then the final score must be of the form $k + 2$ to k for $k \geq n - 1$, and thus an even number of contests will have been played, so again the first remark applies.

(b) As above, a player wins a competition here if and only if that player is ahead when all $2n + 1$ contests are played. Because of the independence assumption, the probability of scores involving $2n + 1$ points depends only on the number of each type of contests played, and not the order. $\square$

Examples of competitions using scoring system (1) are tennis with the classical rules (first to win 6 games by 2) and tennis with a 12 point tiebreaker (this is actually a combination of two competitions of type (1) where the 12 point tiebreaker is invoked if the game score reaches 6-6). Thus in these competitions the probability of winning is independent of which player serves first.

Examples of competitions using the second scoring system are the 9 point tiebreaker in tennis, the World Series in baseball and the playoffs in basketball and ice hockey. The order of points played in a 9 point tiebreaker is 2-2-2-3. The order of game types in the World Series is 2-3-2 while in the other two sports it is 2-2-1-1-1. By the theorem, however, the probability of a person or team winning such competitions is independent of the order used.

2. The Preliminary Process For a Tennis Set

For the rest of this paper we will let

$$p_A = \Pr(\text{player } A \text{ wins a point} \mid \text{player } A \text{ is serving})$$

$$p_B = \Pr(\text{player } B \text{ wins a point} \mid \text{player } B \text{ is serving})$$

$$g_A = \Pr(\text{player } A \text{ wins a game} \mid \text{player } A \text{ is serving})$$

$$g_B = \Pr(\text{player } B \text{ wins a game} \mid \text{player } B \text{ is serving})$$

Our primary assumption is that the probabilities p_A and p_B remain fixed throughout a set or match. Various authors have considered how a player's serving efficacy may be computed under different strategies. They consider two classes of serves—a 'risky' serve that has a high probability of winning the point given that the serve is good, and a 'safe' serve that gives a reduced probability for winning the point, but has a larger probability of being good. We refer the reader to Gale (1971), George (1973) and Gillman (1985) for more details on the calculation of p_A and the best strategies to use. From Kemeny and Snell (1960) we have that

$$g_i = \begin{cases} \frac{1}{2} & \text{if } p_i = \frac{1}{2} \\ \frac{p_i^4(1 - 16(1 - p_i)^4)}{p_i^4 - (1 - p_i)^4} & \text{if } p_i \neq \frac{1}{2} \end{cases} \quad i = A, B. \quad (1)$$

We are interested in computing the probability that player A will win the set given the probabilities above for winning individual points. The preliminary process for a set consists of the states leading to scores of 6-0, 6-1, 6-2, 6-3, 6-4 or 7-5. A *tiebreaker*

occurs when the score reaches 6-6. By a tiebreaker here we will mean either that the players continue playing regular games until one has a two game margin, or a single additional game is played consisting of a 9 or 12 point tiebreaker.

By the theorem we may assume that player *A* serves the first game of the set. Fig. 1 shows the beginning and end states of the preliminary process leading either to outright victory for one player or to a tiebreaker.

Now consider the case that *A* wins by a score 6-*k*, for *k* = 0, 2 or 4. Write *k* = 2*m*. Because an even number of games is played, player *A* must win the last game on his opponent's serve, which occurs with probability 1 - *g_B*. Of the remaining 2*m* + 5 games, the *k* victories for player *B* may be split between the *m* + 3 games in which *A* serves and the *m* + 2 games in which *B* serves. The probability that *B* wins *i* games from the *m* + 3 games served by *A* is binomial with 'success' probability 1 - *g_A*, while the probability that *B* wins the remaining *k* - *i* games when he serves is binomial with 'success' probability *g_B*. Thus the probability that *A* wins by a score 6-0, 6-2 or 6-4 given that *A* serves first is given by

$$(1 - g_B) \sum_{m=0}^2 \sum_{i=0}^{2m} \text{Bin}(m + 3, i, 1 - g_A) \text{Bin}(m + 2, 2m - i, g_B) \tag{2}$$

where Bin(*n*, *i*, *p*) is the binomial probability for *i* successes in *n* trials with success

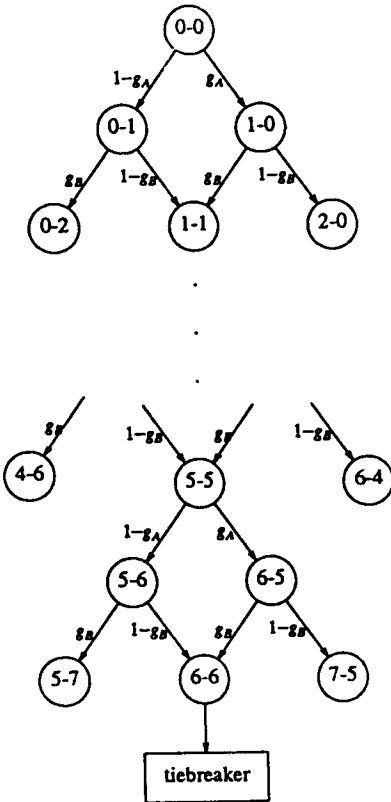


Fig. 1. Preliminary process, player *A* serves first

probability p . A similar argument easily establishes that the probability that A wins by a score of 6-1 or 6-3 is given by

$$g_A \sum_{m=0}^1 \sum_{i=0}^{2m+1} \text{Bin}(m+3, i, 1-g_A) \text{Bin}(m+3, 2m+1-i, g_B). \quad (3)$$

To reach the score 7-5, it is necessary first to attain the score 5-5, after which A must win his serve and then break B 's serve. Thus the probability of a 7-5 score is

$$g_A(1-g_B) \sum_{i=0}^5 \text{Bin}(5, i, 1-g_A) \text{Bin}(5, 5-i, g_B). \quad (4)$$

By summing equations (2), (3) and (4) we obtain the probability that A wins the set without requiring a tiebreaker. We call this probability $N(p_A, p_B)$.

For a tiebreaker to be necessary, the score must first reach 5 games all, followed either by both players holding serve, or both players having their serve broken. Therefore the probability of a tiebreaker is given by

$$T(p_A, p_B) = [g_A g_B + (1-g_A)(1-g_B)] \sum_{i=0}^5 \text{Bin}(5, i, 1-g_A) \text{Bin}(5, 5-i, g_B) \quad (5)$$

Next we consider models for the three tiebreaker procedures.

3. The Tiebreaker Systems

The classical rules of tennis as modified after the 1877 Wimbledon Championships call for the set to continue until one player has a two game margin. This can be modelled as a 5 state random walk with two end absorbing states, as illustrated in Fig. 2. Again we assume that player A serves first. This means that the process starts in state 2 (tie) with A serving. Note that the Markov chain has different probabilities for right and left steps that depend on the particular state. This corresponds to the alternating service pattern.

To compute the probability that the process is absorbed in the state 'set A ', that is, that player A wins the tiebreaker, we consider a first step analysis. Let u_i denote the probability of absorption in state 4 given that the process is currently in state i , for $i = 1, 2, 3$. Then u_1, u_2 and u_3 satisfy the following system of equations derived by considering the possibilities that can arise at the end of the first transition:

$$u_1 = g_B \cdot 0 + (1-g_B) \cdot u_2$$

$$u_2 = (1-g_A) \cdot u_1 + g_A \cdot u_3$$

$$u_3 = g_B \cdot u_2 + (1-g_B) \cdot 1$$

Since the process must start in state 2, we need only solve for u_2 to obtain the probability that A wins the set given that a tiebreaker is played. This probability is

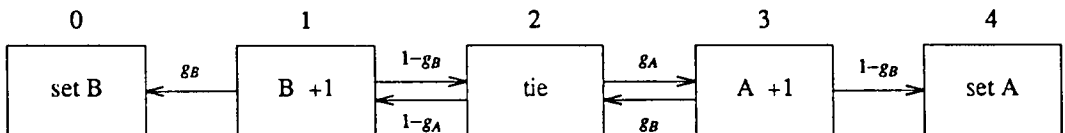


Fig. 2. 5 state random walk for the classical rules

easily found to be

$$u_2 = \frac{g_A(1 - g_B)}{g_A + g_B - 2g_A g_B}. \quad (6)$$

By conditioning on the occurrence of a tiebreaker, we derive the following expression for the probability that A wins the set under the classical rules of tennis.

$$\begin{aligned} S_C(p_A, p_B) &= \Pr(A \text{ wins with no tiebreaker}) + \Pr(A \text{ wins with tiebreaker}) \\ &= \Pr(A \text{ wins, no tiebreaker}) \\ &\quad + \Pr(A \text{ wins tiebreaker} \mid \text{tiebreaker})\Pr(\text{tiebreaker}) \\ &= N(p_A, p_B) + T(p_A, p_B) \left[\frac{g_A(1 - g_B)}{g_A + g_B - 2g_A g_B} \right] \end{aligned} \quad (7)$$

Next we consider the case of a 12 point tiebreaker. As before we assume that player A has served the first game of the set. Then A also serves the first point of the tiebreaker. Player B , then player A , thereafter alternate serving two points each until one player has won at least 7 points by at least a two point margin. This tiebreaker system can be modelled in a similar fashion to that for the classical rules. There is a preliminary stage resulting either in outright victory by a score of 8-6 or 7- k , for $0 \leq k \leq 5$, or resulting in a tied score of 7-7 followed by an 8 state Markov chain with two absorbing states as illustrated in Fig. 3. The Markov chain would start in state 2 since B serves the 15th point of the tiebreaker.

The order of serve for the first 12 points of the tiebreaker is $ABBAABBAABBA$. Let S denote the vector $\langle 3, 3, 4, 5, 5, 5, 6 \rangle$. Then $S(k)$ is the number of times that A serves among the first $6 + k$ points, $0 \leq k \leq 6$. Let W denote the vector $\langle 0, 1, 1, 0, 0, 1 \rangle$, so that

$$W(k) = \begin{cases} 1 & \text{if } A \text{ serves game } 7 + k \\ 0 & \text{if } B \text{ serves game } 7 + k. \end{cases}$$

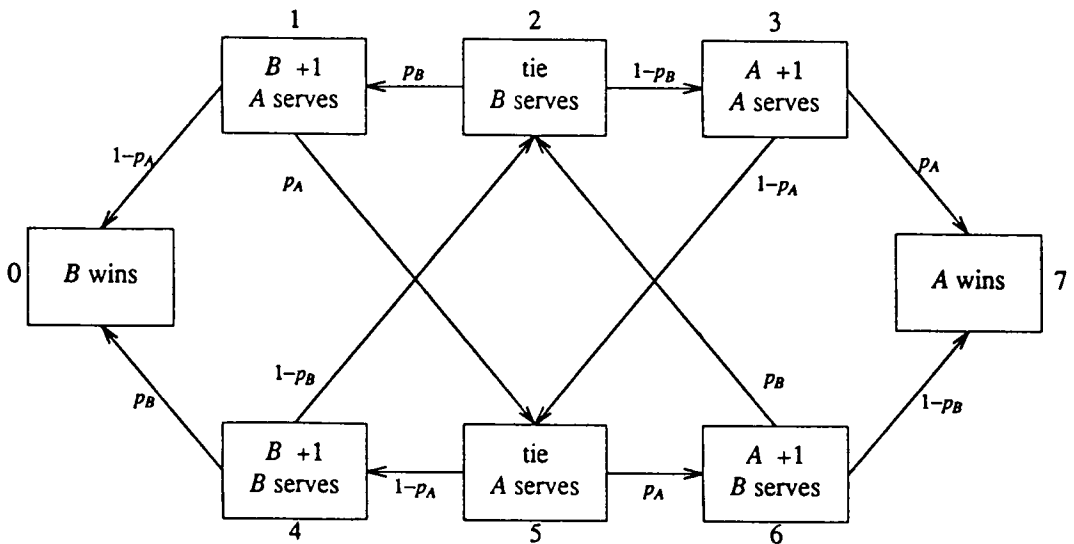


Fig. 3. 8 state random walk for the 12 point tiebreaker

To win the tiebreaker by a score of $7-k$, $0 \leq k \leq 5$, A must win the last point, which occurs with probability p_A if A is serving and with probability $1 - p_B$ if B is serving, and the k points that B wins must be split among the remaining $S(k)$ points that A serves and the $6 + k - S(k)$ points that B serves. Thus the probability that A wins the tiebreaker by a score of $7-k$ for $0 \leq k \leq 5$ is

$$[W(k)p_A + (1 - W(k))(1 - p_B)] \sum_{i=0}^k \text{Bin}(S(k), i, 1 - p_A) \text{Bin}(6 + k - S(k), k - i, p_B)$$

Similarly, the probability that A wins the tiebreaker by a score of $8-6$ is given by

$$p_A(1 - p_B) \sum_{i=0}^6 \text{Bin}(6, i, 1 - p_A) \text{Bin}(6, 6 - i, p_B).$$

since first the score must reach $6-6$ and then A must win the next two points (one served by A and one served by B).

One enters the 8 state Markov chain if the score reaches $7-7$. This occurs with probability

$$[p_A p_B + (1 - p_A)(1 - p_B)] \sum_{i=0}^6 \text{Bin}(6, i, 1 - p_A) \text{Bin}(6, 6 - i, p_B).$$

One may compute the probability of winning the tiebreaker using a first step analysis as with the classical rules. Alternatively, one can make use of the theorem to simplify the computations. Let x denote the probability of absorption in state 7 given that the process starts in state 2, that is,

$$x = \Pr(A \text{ wins} | \text{tie and } B \text{ serving})$$

Then after the first two points, the process has either been absorbed in state 0 or 7, or is in state 5. Thus

$$\begin{aligned} x &= (1 - p_B)p_A + (p_B p_A + (1 - p_B)(1 - p_A))\Pr(A \text{ wins} | \text{tie and } A \text{ serving}) \\ &= (1 - p_B)p_A + (p_B p_A + (1 - p_B)(1 - p_A)) \cdot x \end{aligned}$$

from which we readily deduce that

$$x = \frac{p_A(1 - p_B)}{p_A + p_B - 2p_A p_B}. \quad (8)$$

Thus the probability that A wins a set using a 12 point tiebreaker is

$$\begin{aligned} S_{12}(p_A, p_B) &= \sum_{k=0}^5 [W(k)p_A + (1 - W(k))(1 - p_B)] \\ &\quad \times \sum_{i=0}^k \text{Bin}(S(k), i, 1 - p_A) \text{Bin}(6 + k - S(k), k - i, p_B) \\ &\quad + p_A(1 - p_B) \sum_{i=0}^6 \text{Bin}(6, i, 1 - p_A) \text{Bin}(6, 6 - i, p_B) \\ &\quad + [p_A p_B + (1 - p_A)(1 - p_B)] \sum_{i=0}^6 \text{Bin}(6, i, 1 - p_A) \text{Bin}(6, 6 - i, p_B) \\ &\quad \times \left[\frac{p_A(1 - p_B)}{p_A + p_B - 2p_A p_B} \right]. \end{aligned} \quad (9)$$

Assuming once again that A has served the first game of the set, then A would also serve the first two points of a 9 point tiebreaker. Each player would alternate serving two points until one player wins 5 points. If the score reaches 4 points all, then player B serves the last and deciding point. To compensate for this advantage, the receiver is allowed to choose whether to receive serve in the deuce court or the ad court. This option is not relevant to our model, however.

The order of serve during the tiebreaker is $AABBAABBB$. Following the notation used previously, we let the vector $S = \langle 2, 3, 4, 4, 4 \rangle$ represent the number of points that A serves during the first $4 + k$ points and the vector $W = \langle 1, 1, 0, 0, 0 \rangle$ represent which player serves point $5 + k$, for $k = 0, 1, 2, 3$, or 4 . Then the probability of A winning the 9 point tiebreaker is

$$\sum_{k=0}^4 [W(k)p_A + (1 - W(k))(1 - p_B)] \sum_{i=0}^k \text{Bin}(S(k), i, 1 - p_A) \text{Bin}(4 + k - S(k), k - i, p_B). \quad (10)$$

If we denote the expression in equation (10) by $TIE_9(p_A, p_B)$, then we can write the probability of A winning a set that uses a 9 point tiebreaker as

$$N(p_A, p_B) + T(p_A, p_B)TIE_9(p_A, p_B).$$

The 9 point tiebreaker probability, unlike the previous cases considered, is not symmetric about the line $p_A + p_B = 1$. As an extreme case, consider the situation when $p_A = p_B = 1$, so that both players always win the point on their serve. Then the set will be decided by the winner of the ninth point of the tiebreaker, which is the player that receives the serve in the first game of the set. Thus if p_A is greater than $\frac{1}{2}$, it is to player A 's advantage to serve the second game of the set, not the first, although this advantage only becomes a factor if the set goes into the tiebreaker since the probability of the preliminary process is independent of which player serves first.

4. Probabilities for a Match

The probabilities of winning a set using either the classical rules or the 12 point tiebreaker depend only on the relative abilities of the players and not on which player serves first in the set. Thus it is a simple matter to compute the probability of winning a match consisting of either the best 2 out of 3 sets or the best 3 out of 5 sets. The 9 point tiebreaker system poses a problem, however, for there the probability of winning a set does depend on which player serves the first game, and this in turn depends on which player served first in the preceding set and the score of that set. For example, if A serves first in one set and wins that set in an odd number of games, then B will serve first in the next set. The tiebreaker is counted as one game in these considerations.

Consider the following events for an individual set:

$A_{00} = A$ serves first and A wins in an even number of games

$A_{01} = A$ serves first and A wins in an odd number of games

$A_{10} = B$ serves first and A wins in an even number of games

$A_{11} = B$ serves first and A wins in an odd number of games

$B_{00} = A$ serves first and B wins in an even number of games

$B_{01} = A$ serves first and B wins in an odd number of games

$B_{10} = B$ serves first and B wins in an even number of games

$B_{11} = B$ serves first and B wins in an odd number of games

The first subscript indicates which player served the first game of the set (0 for A and 1 for B) while the second subscript indicates the parity of the number of games (0 for *even* and 1 for *odd*). Thus the transition from one set to the next is represented by the diagram

$$01 \rightarrow 1^*$$

$$00 \rightarrow 0^*$$

$$11 \rightarrow 0^*$$

$$10 \rightarrow 1^*$$

which can be more easily expressed as $(i, j) \rightarrow i + j$, where the addition is done modulo 2. Now let a_{ij} and b_{ij} denote the probability of events A_{ij} and B_{ij} respectively, using the 9 point tiebreaker scoring system. These probabilities can be computed using formula (2) for the even number of games and formulae (3)–(5) and (10) for the odd number of games, with the appropriate substitutions depending on which player serves the first game. Define the matrices P and $\hat{P}$ to be

$$P = \begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix} \quad \hat{P} = \begin{bmatrix} a_{00} & a_{01} \\ a_{11} & a_{10} \end{bmatrix}.$$

Notice that $\hat{P}_{ij} = P_{i, i+j}$ for $i, j = 0, 1$ where again the addition is modulo 2. Let Q and $\hat{Q}$ denote the corresponding matrices for the probabilities b_{ij} for player B .

Routine calculations now show that for a match consisting of the best 2 out of 3 sets

$$\Pr(A \text{ wins } 2/3 \text{ match} | A \text{ serves first}) = [PP + P\hat{Q}P + Q\hat{P}P]_{0^*},$$

and that for a match consisting of the best 3 out of 5 sets, the probability that A wins the match given that A serves first is

$$\begin{aligned} & [P\hat{P}P + P\hat{P}\hat{Q}P + P\hat{Q}\hat{P}P + Q\hat{P}\hat{P}P + P\hat{P}\hat{Q}\hat{Q}P + P\hat{Q}\hat{Q}\hat{P}P + Q\hat{Q}\hat{P}\hat{P}P \\ & + P\hat{Q}\hat{P}\hat{Q}P + Q\hat{P}\hat{P}\hat{Q}P + Q\hat{P}\hat{Q}\hat{P}P]_{0^*}. \end{aligned}$$

where the subscript 0^* denotes summation over the first row.

5. Numerical Examples

In this section we present numerical results for six cases: $p_A = 0.5, 0.6, 0.7, 0.8, 0.9$ and 1.0 . In each case we let p_B range from 0.0 to 1.0 . Probabilities for the following quantities are displayed in Table 1—(1) the probability of a tiebreaker, (2) the probability of A winning the set under the classical rules, (3) the probability of A winning the set under a 12 point tiebreaker, and (4) the probabilities of A winning the set under a 9 point tiebreaker when A serves first and then when B serves first. The last column of each section gives the unconditioned probability $S_9(p_A, p_B)$ of A winning the set with a 9 point tiebreaker, computed by averaging the probabilities of A winning given A serves first and given B serves first. This reflects the practice of

using a coin (or racket) flip to determine order of serve; the player winning the toss has the option of serving first or second.

It is quickly evident from looking at Table 1 that a small advantage in winning a point when serving becomes magnified when viewed in terms of winning the set. For two players of equal ability ($p_A = p_B$) the probability of either player winning the set under any tiebreaker system is 0.5, as agrees with our intuition. On the other hand, if $p_A = 0.7$ and $p_B = 0.6$ then this advantage for A yields a probability of 0.81 for A winning the set using the classical scoring rules. Even when $p_B = 0.69$ player A 's probability of winning the set increases from 0.50 to 0.54. Of course, seldom do we actually know the values of p_A and p_B . Rather, we use a scoring system in a competition so as to decide, on the basis of the outcome of the competition, which is the better

TABLE 1
Probabilities of A winning a set under various scoring systems for given p_A and p_B

$p_A = 0.5$							$p_A = 0.6$						
p_B	Tiebreaker	S_C	S_{12}	S_{9A}	S_{9B}	S_9	p_B	Tiebreaker	S_C	S_{12}	S_{9A}	S_{9B}	S_9
0.0	0.016	1.	1.	1.	1.	1.	0.0	0.	1.	1.	1.	1.	1.
0.1	0.016	1.	1.	0.999	0.998	0.999	0.1	0.	1.	1.	1.	1.	1.
0.2	0.022	0.995	0.993	0.993	0.991	0.992	0.2	0.	1.	1.	1.	1.	1.
0.3	0.047	0.963	0.958	0.956	0.954	0.955	0.3	0.005	0.996	0.996	0.996	0.996	0.996
0.4	0.094	0.820	0.813	0.811	0.808	0.810	0.4	0.027	0.966	0.963	0.962	0.962	0.962
0.5	0.123	0.500	0.500	0.500	0.500	0.500	0.5	0.094	0.820	0.813	0.808	0.811	0.809
0.6	0.094	0.180	0.187	0.189	0.192	0.190	0.6	0.174	0.500	0.500	0.495	0.505	0.500
0.7	0.047	0.037	0.042	0.044	0.046	0.045	0.7	0.190	0.186	0.205	0.204	0.220	0.212
0.8	0.022	0.005	0.007	0.007	0.009	0.008	0.8	0.169	0.038	0.059	0.062	0.079	0.070
0.9	0.016	0.	0.	0.001	0.002	0.001	0.9	0.159	0.002	0.012	0.015	0.031	0.023
1.0	0.016	0.	0.	0.	0.	0.	1.0	0.159	0.	0.	0.	0.012	0.006
$p_A = 0.7$							$p_A = 0.8$						
p_B	Tiebreaker	S_C	S_{12}	S_{9A}	S_{9B}	S_9	p_B	Tiebreaker	S_C	S_{12}	S_{9A}	S_{9B}	S_9
0.0	0.	1.	1.	1.	1.	1.	0.0	0.	1.	1.	1.	1.	1.
0.1	0.	1.	1.	1.	1.	1.	0.1	0.	1.	1.	1.	1.	1.
0.2	0.	1.	1.	1.	1.	1.	0.2	0.	1.	1.	1.	1.	1.
0.3	0.	1.	1.	1.	1.	1.	0.3	0.	1.	1.	1.	1.	1.
0.4	0.005	0.996	0.996	0.996	0.996	0.996	0.4	0.	1.	1.	1.	1.	1.
0.5	0.047	0.963	0.958	0.954	0.956	0.955	0.5	0.230	0.995	0.993	0.991	0.993	0.992
0.6	0.190	0.814	0.795	0.780	0.796	0.788	0.6	0.169	0.962	0.941	0.921	0.938	0.930
0.7	0.381	0.500	0.500	0.477	0.523	0.500	0.7	0.498	0.842	0.768	0.706	0.783	0.745
0.8	0.498	0.158	0.232	0.217	0.294	0.255	0.8	0.778	0.500	0.500	0.416	0.584	0.500
0.9	0.532	0.012	0.078	0.076	0.169	0.123	0.9	0.869	0.061	0.238	0.184	0.425	0.304
1.0	0.534	0.	0.	0.	0.090	0.045	1.0	0.876	0.	0.	0.	0.287	0.144
$p_A = 0.9$							$p_A = 1.0$						
p_B	Tiebreaker	S_C	S_{12}	S_{9A}	S_{9B}	S_9	p_B	Tiebreaker	S_C	S_{12}	S_{9A}	S_{9B}	S_9
0.0	0.	1.	1.	1.	1.	1.	0.0	0.	1.	1.	1.	1.	1.
0.1	0.	1.	1.	1.	1.	1.	0.1	0.	1.	1.	1.	1.	1.
0.2	0.	1.	1.	1.	1.	1.	0.2	0.	1.	1.	1.	1.	1.
0.3	0.	1.	1.	1.	1.	1.	0.3	0.	1.	1.	1.	1.	1.
0.4	0.	1.	1.	1.	1.	1.	0.4	0.	1.	1.	1.	1.	1.
0.5	0.016	1.	1.	0.998	0.999	0.999	0.5	0.016	1.	1.	1.	1.	1.
0.6	0.159	0.998	0.988	0.969	0.985	0.977	0.6	0.159	1.	1.	0.988	1.	0.994
0.7	0.532	0.988	0.922	0.831	0.924	0.877	0.7	0.534	1.	1.	0.910	1.	0.955
0.8	0.869	0.939	0.762	0.575	0.816	0.696	0.8	0.876	1.	1.	0.713	1.	0.856
0.9	0.983	0.500	0.500	0.296	0.704	0.500	0.9	0.991	1.	1.	0.415	1.	0.707
1.0	0.991	0.	0.	0.	0.585	0.293	1.0	1.	*	*	0.	1.	0.500

Notes: (1) '0.' indicates < 0.001 ; '1.' indicates > 0.999
(2) * indicates that set extends indefinitely

player. The 'ideal' scoring system, therefore, would be one that maximised the probability of victory for the player A with $p_A > p_B$. Figs 4 and 5 illustrate the probabilities of victory in a set for player A when $p_A = p_B + \varepsilon$, for $\varepsilon = 0.1$ and $\varepsilon = 0.01$ respectively, for each of the three scoring systems already discussed, as well as for the Van Alan No-Ad system in which the first player to win 4 points wins the game, with a 9 point tiebreaker played at 6-6.

The probability $S_{VA}(p_A, p_B)$ of A winning a No-Ad set (averaged over the two cases that A serves the first game and that B serves the first game) may be computed exactly as for the 9 point tiebreaker system with the only change occurring in the computation of g_A and g_B . Here we need only compute the probability of A (or B) winning 4 out of 7 points. The appropriate probability is

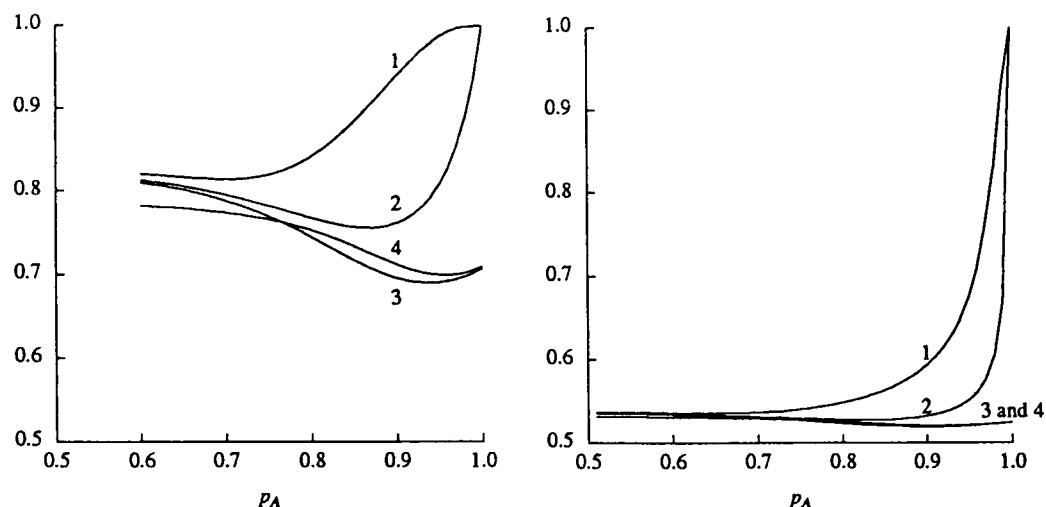
$$\hat{g}_i = \binom{3}{0}p_i^4 + \binom{4}{1}p_i^4(1-p_i) + \binom{5}{2}p_i^4(1-p_i)^2 + \binom{6}{3}p_i^4(1-p_i)^3$$

for $i = A$ or B .

We see from Figs 4 and 5 that the classical rules are optimal among the four scoring systems discussed. The disadvantage of this system is that the expected length of the set grows as p_A tends to 1 (although in practice such large probabilities are seldom encountered). If one restricts attention to scoring systems with a fixed upper bound on the possible number of points, then the 9-point tiebreaker is optimal for values of p_A less than 0.75 (approximately), which is a range appropriate for most tennis competitions.

6. Goodness-of-fit Tests

Our basic assumption in the preceding work has been that the probabilities p_A and p_B remain fixed throughout a set. This is obviously unrealistic in practice as weather conditions, momentum and fatigue, among other factors, all contribute to a player's service effectiveness. Over the course of many games and sets, however, such factors might even out so that the assumption of a fixed probability becomes



Figs 4 and 5. Probability of A winning set for $p_A = p_B + \varepsilon$ for $\varepsilon = 0.1, 0.01$: (1) classical rules (2) 12 point tiebreaker (3) 9 point tiebreaker (4) No-Ad scoring

reasonable. We present here empirical evidence for such a claim. Tables 2 and 3 contain the results of 4 years of ‘friendly’ competition between Professors John D’Angelo and John Marden of the Departments of Mathematics and Statistics, respectively, at the University of Illinois, and 9 years of competition between Chris Evert and Martina Navratilova in the finals of professional tournaments.

Unfortunately, as is the case with almost all records of tennis competition, the determination of p_A and p_B for the players involved is not possible. We also do not know which games were won while serving and thus are forced to assume that $p_B = 1 - p_A$. For the D’Angelo–Marden competition we take D’Angelo as player A . From the data we estimate for each year the probability that D’Angelo wins a game to be 0.609, 0.605, 0.604 and 0.620 respectively, with an overall estimate of $p_A = 0.609$ for the 4496 games played during the four years. We use these estimates to test two aspects of the data: (i) the fit for the individual set scores (where the last 4 classes have been combined), and (ii) the fit for the number of sets won by D’Angelo each year. The observed and expected values are displayed in Table 2. The χ^2 value for (i) is 13.93 with 5 degrees of freedom, which gives a p value between 0.01 and 0.025 ($\chi^2_{5,0.975} = 12.83$, $\chi^2_{5,0.99} = 15.09$). Thus at a 0.05 level of significance we would accept the null hypothesis that the observed values fit the model. The χ^2 value for (ii) is 0.34 with 2 degrees of freedom, which has a p value between 0.975 and 0.99.

From the second table we take Navratilova as player A and estimate her probability of winning a game as $g_A = 0.567$. Using this value in equation (1), we estimate $p_A = 0.527$. Finally, from equation (9) we obtain the theoretic value of 0.686 for Navratilova to win a set using a 12 point tiebreaker. Among the 100 sets played between the two players, Navratilova actually won 68%. The observed and expected frequency of the number of each set score is also displayed in Table 3. They yield a χ^2 value of 5.82 with 5 degrees of freedom and a p value between 0.30 and 0.40. Again we seem to have a good fit.

TABLE 2
Scores for D’Angelo–Marden (classical rules)

score	1983	1984	1985	1986	total sets	expected sets
6-0	9-1	12-0	10-2	4-0	40	25.58
6-1	18-3	14-7	11-2	18-4	77	64.64
6-2	16-5	22-5	16-3	21-2	90	93.32
6-3	8-7	22-6	28-6	17-1	95	105.18
6-4	15-6	23-8	20-10	15-8	105	103.32
7-5	5-3	9-5	10-4	7-5	48	49.21
8-6	5-1	2-3	5-1	5-1	32	44.74
9-7	2-0	1-0	1-1	4-0		
10-8	0-0	1-0	0-0	0-0		
11-9	0-0	1-0	0-0	0-0		
games	568-364	795-518	728-477	649-397		
games%	0.609	0.605	0.604	0.620		
sets	78-26	107-34	101-29	91-21	487	486.99
sets%	0.750	0.759	0.777	0.812		
from model	0.792	0.783	0.781	0.815		
expected sets for D’Angelo	82.37	110.4	101.53	91.28		

Source: personal communication

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TABLE 3
Scores for Navratilova-Evert (12 point tiebreaker)

score	1978-1987	total sets	expected sets
6-0	4-2	6	3.26
6-1	9-2	11	9.62
6-2	15-3	18	16.60
6-3	13-11	24	21.87
6-4	16-9	25	24.39
7-5	6-3	9	12.16
7-6	5-2	7	12.11
games	523-399		
games%	0.567		
sets	68-32	100	100.01
sets%	0.680		
from model	0.686		

Source: Sports Illustrated (For the Record)

7. Conclusion

Morris (1977) has studied models of scoring systems in which on specific points one of the players is able to increase his chance of winning by some fixed factor, called the *importance* of that point. A sudden death match-point has maximal importance. The data presented here, however, indicate that our simple models may well serve a useful purpose in comparing the effectiveness of scoring systems to discriminate between two players.

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